Audit, recovery, and trace logging in SnapDrive for UNIX

Snapdrive for Unix

NetApp
August 06, 2021

This PDF was generated from https://docs.netapp.com/us-en/snapdrive-unix/aix/concept_types_of_logs.html on August 06, 2021. Always check docs.netapp.com for the latest.

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Audit, recovery, and trace logging in SnapDrive for UNIX

SnapDrive for UNIX helps you maintain various types of log files such as audit, recovery, and trace log files. These log files are used later for troubleshooting.

Types of logs

SnapDrive for UNIX supports various types of log files, which help you troubleshoot when SnapDrive for UNIX is not working as expected.

- Audit log

  SnapDrive for UNIX logs all commands and their return codes in an audit log. SnapDrive for UNIX makes an entry when you initiate a command and another when the command is complete. The later entry includes both the status of the command and the completion time.

- Recovery log

  Some SnapDrive for UNIX operations have the potential to leave the system in an inconsistent or less usable state if interrupted. This situation could occur if a user terminates the program, or if the host crashes in the middle of an operation. The recovery log contains the steps of a Snap restore operation. It documents the steps that are taken and the progress made so that technical support can assist you with the manual recovery process.

- Trace log

  SnapDrive for UNIX reports information useful for diagnosing problems. If you have a problem, NetApp technical support might request this log file.

  SnapDrive 4.0 for UNIX introduced SnapDrive for UNIX daemon service. For SnapDrive for UNIX commands to run, the daemon must be running. Daemon tracing must be carried out on for each individual command, even when commands are run in parallel. The location and name of the log file is `/var/log/sd-trace.log`. These trace files have only the command-specific trace entries. The trace logs for daemon execution are recorded in a different trace file `/var/log/sd-daemon-trace.log`.

Enabling and disabling log files

If you want to enable a log file, specify a file name as the value in the name-value pair of the log file you want to enable. If you want to disable a file, do not enter a value for the log file name parameter.

You must have logged in as a root user.

Steps

1. Open the `snapdrive.conf` file in a text editor.
2. Choose the appropriate option to either enable or disable a log file.

If you want to…	Then…
Enable a log file	Specify a file name as the value in the name-value pair of the log file you want to enable. SnapDrive for UNIX only writes log files if it has the name of a file to write to. The default names for the log files are as follows: • Audit log: `sd-audit.log` • Recovery log: `sd-recovery.log` • Trace log: `sd-trace.log` The path to these files might vary depending on your host operating system.
Disable a log file	Do not enter a value for the log file name parameter. If you do not supply a value, there is no file name to which SnapDrive for UNIX can write the log information. **Example** This example disables the audit log file. `audit-log-file=""`

3. Save the `snapdrive.conf` file after you make all your changes.

   SnapDrive for UNIX automatically checks this file each time it starts. You must restart the SnapDrive for UNIX daemon for the changes to take effect.

Log file rotation settings

The values you specify for the variable in the `snapdrive.conf` file affect the log file rotation.

The values you specify in the `snapdrive.conf` file enable automatic log file rotations. You can change these values, if necessary, by editing the `snapdrive.conf` variables. The following options affect log file rotation:

- `audit-log-max-size`
- `audit-log-save`
- `trace-max-size`
- `trace-log-max-save`
- `recovery-log-save`

With automatic log rotation, SnapDrive for UNIX keeps old log files until it reaches the limit specified in the `audit-log-save`, `trace-log-save`, and `recovery-log-save` variable. Then it deletes the oldest log file.

SnapDrive for UNIX tracks which file is oldest by assigning the file the number "0" when it creates the file. Each

time it creates a new file, it increments by 1 the number assigned to each of the existing log files. When a log file's number reaches the save value, SnapDrive for UNIX deletes that file.

Example: This example uses the `ls` command to display information about the log files on the system.

Based on those settings, you would see the following information in log files.

```
# ls -l /var/log/sd*
-rw-r--r-- 1 root other 12247 Mar 13 13:09 /var/log/sd-audit.log
-rw-r--r-- 1 root other 20489 Mar 12 16:57 /var/log/sd-audit.log.0
-rw-r--r-- 1 root other 20536 Mar 12 03:13 /var/log/sd-audit.log.1
-rw-r--r-- 1 root other 3250 Mar 12 18:38 /var/log/sd-recovery.log.1
-rw-r--r-- 1 root other 6250 Mar 12 18:36 /var/log/sd-recovery.log.2
-rw-r--r-- 1 root other 6238 Mar 12 18:33 /var/log/sd-recovery.log.3
-rw-r--r-- 1 root other 191704 Mar 13 13:09 /var/log/sd-trace.log
-rw-r--r-- 1 root other 227929 Mar 12 16:57 /var/log/sd-trace.log.0
-rw-r--r-- 1 root other 213970 Mar 12 15:14 /var/log/sd-trace.log.1
-rw-r--r-- 1 root other 261697 Mar 12 14:16 /var/log/sd-trace.log.2
-rw-r--r-- 1 root other 232904 Mar 12 14:15 /var/log/sd-trace.log.3
-rw-r--r-- 1 root other 206905 Mar 12 14:14 /var/log/sd-trace.log.4
```

Contents of an audit log file

The audit log shows information about commands you issued with SnapDrive for UNIX.

The audit log file maintains the history of the following information:

- The commands issued.
- The return value from those commands.
- The user ID of the user who invoked the command.
- A timestamp indicating when the command started (with no return code) and another timestamp indicating when the command finished (with a return code). The audit log record shows only information about `snapdrive` use (issued commands).

An audit log file contains the following information.

Field	Description
uid	user ID
gid	group ID
msgText	message text
returnCode	return code from a command

Changing the defaults for the audit logs

You can use `snapdrive.conf` file to change the logging parameters of the audit log files such as maximum size of audit log file, and maximum number of old audit files.

The `snapdrive.conf` file enables you to set the following values for audit logging:

- The maximum size of the audit log file. The default size is 20K. After the file size reaches the value specified in the `snapdrive.conf` file, SnapDrive for UNIX renames the current audit log file by adding an arbitrary number to the name. Then, a new audit file is started using the name specified by the `audit-log-file` value.
- The maximum number of old audit files that SnapDrive for UNIX saves. The default is 2.

Example of an audit log file:

```
2501: Begin uid=0 gid=1 15:35:02 03/12/04 snapdrv snap create -dg
rdg -snapname snap_rdg1
2501: Status=0 15:35:07 03/12/04
2562: Begin uid=0 gid=1 15:35:16 03/12/04 snapdrv snap create -dg
rdg -snapname snap_rdg1
2562: FAILED Status=4 15:35:19 03/12/04
```

The first pair of lines in this example shows an operation that succeeded, as indicated by the "Status=0" line.

The second pair of lines indicates an operation that failed. The return code of "4" means “already exists.” If you look at the two command lines, you can see that the first created a Snapshot copy called `snap_rdg1`. The second line attempted to do the same, but the name already existed, so the operation failed.

Contents of the recovery log

The recovery log helps you to trace the reasons for an inconsistent system state. SnapDrive for UNIX generates this file when some operation stops working.

If you press Ctrl-C to halt SnapDrive for UNIX, or if the host or storage system fails in the middle of an operation, the system might not be able to recover automatically.

During any operation that, if interrupted, could leave the system in an inconsistent state, SnapDrive for UNIX writes information to a recovery log file. If a problem occurs, you can send this file to NetApp technical support so they can assist you in recovering the system’s state.

The recovery log utility maintains the records of the commands that are issued during the operation. Each command is marked with an operation_index (a number that uniquely identifies the operation being executed), followed by the date/time stamp and the message text.

Default values for the recovery logs

You can change the name of the recovery log file and the maximum number of old recovery log files that SnapDrive for UNIX saves.

The `snapdrive.conf` file enables you to set the following values for recovery logging:

- The name of the file containing the recovery log, such as `recovery.log`.
- The maximum number of old recovery files that SnapDrive for UNIX saves. The default is 20. SnapDrive for UNIX keeps this number of recovery logs, if the problem with the process is not identified immediately. SnapDrive for UNIX starts a new recovery log file each time it completes an operation. Then, renames the previous one by adding an arbitrary number to the name, such as `recovery.log.0`, `recovery.log.1`, and so on.

> The size of the recovery log file depends on the operation that is performed. Each recovery log contains information about a single operation. When that operation is complete, SnapDrive for UNIX starts a new recovery log, regardless of how large the previous file was. As a result, there is no maximum size for a recovery log file.

The following sample recovery log shows that SnapDrive for UNIX has restored two Snapshot copies before the operations halted. You could send this recovery log file to technical support for assistance in restoring the remaining Snapshot copies.

```
6719: BEGIN 15:52:21 03/09/04 snapdrive snap restore -dg jssdg -
snapname natasha:/vol/vol1:abort_snap_restore
6719: BEGIN 15:52:27 03/09/04 create rollback snapshot:
natasha:/vol/vol1:abort_snap_restore.RESTORE_ROLLBACK_03092004_155
225
6719: END 15:52:29 03/09/04 create rollback snapshot:
natasha:/vol/vol1:abort_snap_restore.RESTORE_ROLLBACK_03092004_155
225 successful
6719: BEGIN 15:52:29 03/09/04 deactivate disk group: jssdg
6719: BEGIN 15:52:29 03/09/04 stop host volume:
/dev/vx/dsk/jssdg/jvol_1
6719: END 15:52:30 03/09/04 stop host volume:
/dev/vx/dsk/jssdg/jvol_1 successful
6719: BEGIN 15:52:30 03/09/04 unmount file system: /mnt/demo_fs
6719: END 15:52:30 03/09/04 unmount file system: /mnt/demo_fs
successful
6719: BEGIN 15:52:30 03/09/04 stop host volume:
/dev/vx/dsk/jssdg/jvol_2
6719: END 15:52:30 03/09/04 stop host volume:
/dev/vx/dsk/jssdg/jvol_2 successful
6719: BEGIN 15:52:30 03/09/04 deport disk group: jssdg
6719: END 15:52:30 03/09/04 deport disk group: jssdg successful
6719: END 15:52:30 03/09/04 deactivate disk group: jssdg
successful
6719: BEGIN 15:52:31 03/09/04 SFSR of LUN: /vol/vol1/lun1 from
snapshot: abort_snap_restore
6719: END 15:52:31 03/09/04 SFSR of LUN: /vol/vol1/lun1 from
snapshot: abort_snap_restore successful
6719: BEGIN 15:52:47 03/09/04 SFSR of LUN: /vol/vol1/lun2 from
snapshot: abort_snap_restore
6719: END 15:52:47 03/09/04 SFSR of LUN: /vol/vol1/lun2 from
snapshot: abort_snap_restore successful
```

What the trace log file is

Technical support uses the trace log file to troubleshoot problems.

Enabling the trace log file does not affect system performance. By default, this file is enabled. You can disable by setting the `snapdrive.conf trace-enabled` variable to `off`.

Default values for the trace log files

The trace log files allow you to set various parameters and also change them. These parameters must be set in the `snapdrive.conf` file.

The following values must be set in the `snapdrive.conf` file:

- The name of the file containing the trace log.
- The maximum size of the trace log file. The default size is "0 " bytes. This value ensures that each trace log file contains only one SnapDrive for UNIX command.

  If you reset the default size to a value other than 0, when the file reaches the size you specified, SnapDrive for UNIX renames the current trace log file by adding an arbitrary number to the name. Then it starts a new trace log file using the name specified by the `trace-log-file` value.

- The maximum number of old trace files that SnapDrive for UNIX saves. The default is 100.
- The types of messages that SnapDrive for UNIX writes to the trace log file. By default, the trace log file contains fatal errors, admin errors, command errors, warnings, and information messages.

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